



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

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22AI503/22AM503 MACHINE LEARNING

UNIT II – DATA PREPROCESSING

Data Objects and Attribute Types

1. Introduction

1.1 Definition

In machine learning and data mining, data is organised into data objects (also called samples, instances, or records) that are described by a set of attributes (also called features, variables, or dimensions). Understanding the nature of these attributes is the first and most critical step before any pre-processing or modelling activity.

Key Definitions:

A Data Object represents a single entity in the dataset (e.g., one customer, one transaction, one patient).

An Attribute is a field or feature that describes a property of the data object.

The collection of attributes for all data objects forms the Feature Matrix used by ML algorithms.

1.2 Importance in Machine Learning

Attribute types determine which preprocessing steps are applicable, which distance or similarity measures are valid, and which ML algorithms can be used directly or after encoding. Misidentifying an attribute type leads to incorrect model assumptions and degraded performance.

1.3 Role in Data Preprocessing Pipeline

Data preprocessing begins by profiling each attribute — determining its type, range, missing-value rate, and distribution. This profile guides subsequent steps: normalization (numeric), encoding (categorical), and binning (continuous to discrete).

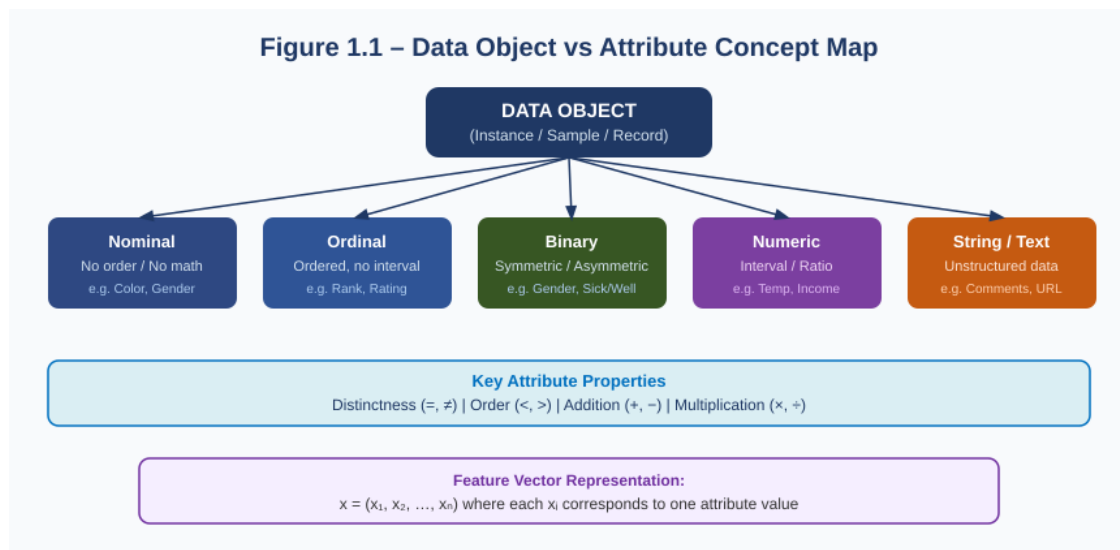


Figure 1.1 – Data Object vs Attribute Concept Map

2. Learning Outcomes

Understand

- The distinction between data objects and attributes
- The four measurement scales: Nominal, Ordinal, Interval, and Ratio
- The difference between symmetric and asymmetric binary attributes

Apply

- Identify attribute types in real-world datasets (Iris, Titanic, medical records)
- Select appropriate preprocessing techniques based on attribute type
- Perform one-hot encoding and label encoding for categorical attributes

Analyze

- Evaluate the impact of wrong attribute-type assumptions on ML models
- Choose suitable distance/similarity metrics based on attribute types
- Detect and handle sparse, high-dimensional attribute spaces

3. Detailed Technical Concepts

3.1 Data Objects

A data object is one row in a dataset table. Each data object is described by attribute values forming a feature vector $x = (x_1, x_2, \dots, x_n)$. A dataset $D = \{x^1, x^2, \dots, x^N\}$ consists of N data objects each with d attributes. Data objects may also be called:

- Samples, examples, instances (in machine learning)
- Observations (in statistics)
- Records, tuples (in databases)
- Points or vectors (in geometry/linear algebra)

3.2 Attribute Types — The Four Measurement Scales

Following the Stevens (1946) classification, attributes are categorised by the mathematical operations that are meaningful on their values:

3.2.1 Nominal Attributes (Categorical)

Nominal attributes represent categories with no inherent ordering. Only equality (=) and inequality (≠) are meaningful.

- Examples: Blood type {A, B, AB, O}, Country name, Hair colour, ZIP code
- Encoding: One-Hot Encoding (OHE) — creates binary dummy variables for each category
- Distance: Hamming distance, Simple Matching Coefficient (SMC)

One-Hot Encoding example for "Color" = {Red, Blue, Green}:

Red → [1, 0, 0]

Blue → [0, 1, 0]

Green → [0, 0, 1]

3.2.2 Ordinal Attributes

Ordinal attributes have a meaningful order, but the intervals between values are not equal or defined.

- Examples: Movie rating {Poor, Fair, Good, Excellent}, Educational level {High School < UG < PG < PhD}
- Encoding: Label/Ordinal Encoding — assigns integers preserving order (e.g., 1, 2, 3, 4)
- Note: Arithmetic operations are not valid. "Excellent – Fair ≠ Good – Poor" in magnitude

3.2.3 Interval Attributes

Interval attributes have meaningful order and equal spacing between values but no true zero point. Addition and subtraction are valid; multiplication and division are not.

- Examples: Temperature in °C/°F, Calendar dates, IQ scores, pH scale
- 20°C is not "twice as hot" as 10°C because 0°C is arbitrary, not absence of temperature
- Preprocessing: Min-Max normalization or Z-score standardization

3.2.4 Ratio Attributes

Ratio attributes have all properties of interval attributes plus a meaningful zero point, making ratio comparisons valid.

- Examples: Height, Weight, Income, Age, Distance, Time duration, Kelvin temperature
- "60 kg is twice as heavy as 30 kg" is a valid statement
- All arithmetic operations (=, ≠, <, >, +, −, ×, ÷) are meaningful

3.3 Binary Attributes

Binary attributes are a special case with only two states (0 and 1). They are further classified:

3.3.1 Symmetric Binary

Both states are equally important. Example: Gender (Male/Female). The Simple Matching Coefficient (SMC) is used:

$$\text{SMC} = (f_{11} + f_{00}) / (f_{11} + f_{01} + f_{10} + f_{00})$$

3.3.2 Asymmetric Binary

One state is more meaningful (usually 1 = positive/rare). Example: Disease test result (1=Positive is rare and important). The Jaccard coefficient is used:

Jaccard = $f_{11} / (f_{11} + f_{01} + f_{10})$ — f_{00} (both negative) is ignored

3.4 Discrete vs Continuous Attributes

An alternative classification orthogonal to the measurement scale:

- Discrete: A finite or countably infinite set of values. Includes all nominal/ordinal attrs. Examples: Number of children, word count in a document.
- Continuous: Real-valued with infinite precision. Typically ratio/interval attrs. Examples: Height 175.36 cm, Temperature 37.2°C. Requires binning/discretization for some algorithms.

Figure 1.2 – Attribute Types Measurement Scale Hierarchy

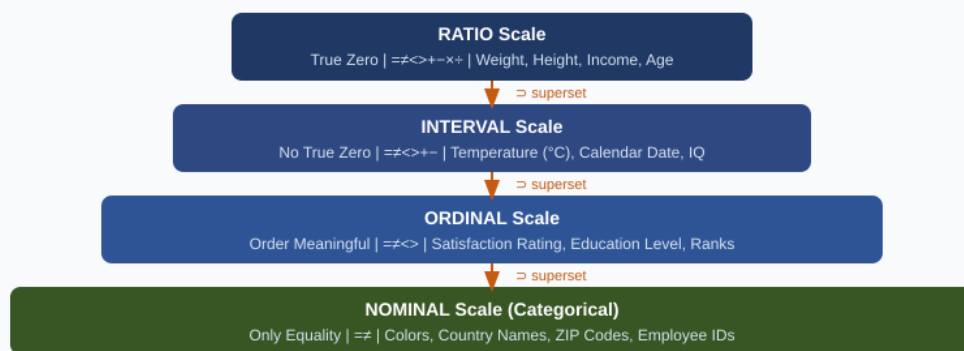


Figure 1.2 – Attribute Types Measurement Scale Hierarchy

3.5 Algorithms and Encoding Techniques

Attribute Type	Operations	Encoding Method	ML Algorithms Compatible
Nominal	=, ≠	One-Hot Encoding / OHE	Naïve Bayes, Decision Tree, kNN (with SMC)
Ordinal	=, ≠, <, >	Label/Ordinal Encoding	Decision Tree, Ordinal Regression, SVM
Interval	=, ≠, <, >, +, -	Z-score / Min-Max	Linear Regression, SVM, kNN (Euclidean)
Ratio	All ops	Log Transform / Z-score	All ML algorithms after normalization
Binary	=, ≠	Keep as 0/1	Logistic Regression, kNN (Jaccard/SMC)

3.6 Investigation Workflow

- Step 1 — Load Dataset: Import CSV/database table into a DataFrame (pandas/R).
- Step 2 — Profile Attributes: Use df.dtypes, df.describe(), df.nunique() to identify attribute types.
- Step 3 — Classify Each Attribute: Assign nominal/ordinal/interval/ratio/binary label.
- Step 4 — Plan Encoding: Select OHE for nominal, label encoding for ordinal, scaling for numeric.
- Step 5 — Validate: After encoding, verify the feature matrix shape and dtype correctness.

4. Architecture / Process Flow

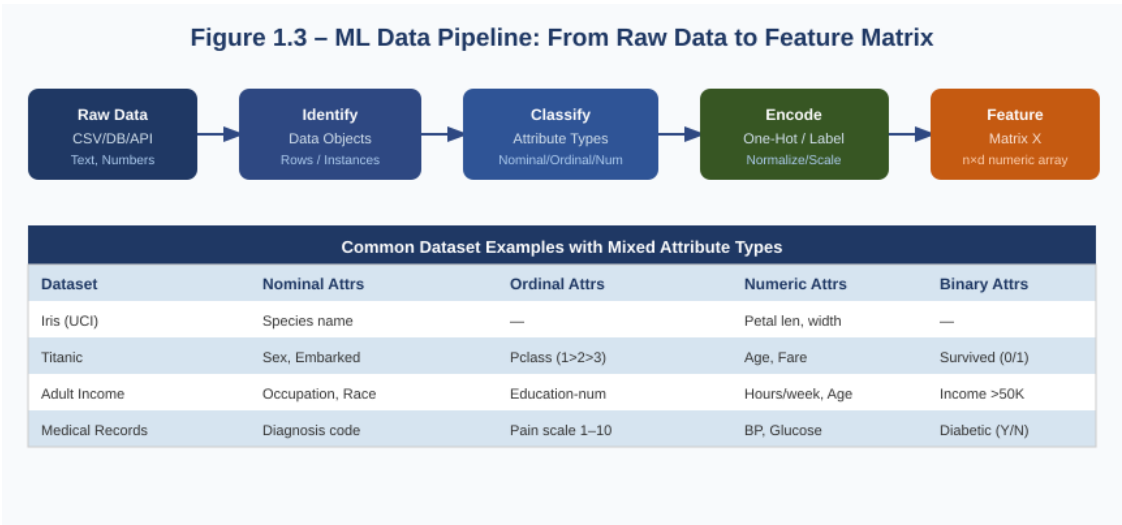


Figure 1.3 – ML Data Pipeline: From Raw Data to Feature Matrix

5. Industry Tools and Technologies

5.1 Commercial Tools

Tool	Vendor	Key Feature	Attribute Type Support
SPSS Modeler	IBM	Visual data profiling pipeline	Auto type detection with metadata
SAS Enterprise Miner	SAS	Automatic attribute role assignment	Nominal, Ordinal, Interval, Binary
Alteryx Designer	Alteryx	Drag-drop type conversion nodes	One-Hot, Label, Date, Numeric
DataRobot	DataRobot	AutoML with auto type inference	Smart encoding selection per type
Talend Studio	Talend	ETL pipeline with schema mapping	Schema validation and type casting

5.2 Open-Source Tools

- pandas (Python): df.dtypes, pd.Categorical, pd.get_dummies() for OHE
- scikit-learn: LabelEncoder, OrdinalEncoder, OneHotEncoder in sklearn.preprocessing

- R base + tidyverse: factor(), as.ordered(), model.matrix() for encoding
- Featuretools: Automated feature engineering respecting attribute semantics
- WEKA: Attribute profiling GUI, nominal-to-numeric filters built-in

6. Real-World Applications

Domain	Data Object	Nominal Attrs	Ordinal Attrs	Ratio Attrs
Healthcare	Patient record	Diagnosis code, Blood type	Pain level (1–10)	Age, BMI, Blood pressure
E-Commerce	Order transaction	Product category, Payment	Customer rating (1–5)	Price, Quantity, Discount
Education	Student result	Department, Course code	Grade (A>B>C>D>F)	Marks (0–100), GPA
Finance	Loan application	Employment type, State	Credit score band	Income, Loan amount
Manufacturing	Sensor reading	Machine ID, Shift type	Defect severity level	Temperature, RPM, Pressure

7. Advantages and Limitations

Advantages of Proper Attribute Classification	Limitations / Challenges
Guides correct choice of distance metrics	Manual classification is time-consuming for large schemas
Prevents invalid mathematical operations on categories	Auto-detection tools may misclassify ZIP codes as ratio
Enables appropriate normalization and encoding	Mixed-type datasets complicate pipeline design
Improves model accuracy and interpretability	Ordinal encoding can imply incorrect interval assumptions
Reduces dimensionality issues from incorrect OHE	High-cardinality nominals cause OHE explosion (1000+ columns)

8. Case Study

CASE STUDY

Case: UCI Adult Income Dataset — Attribute Misclassification Impact

Problem: A data scientist used the "education-num" attribute (ordinal: 1=Preschool to 16=Doctorate) as a raw integer in a linear regression model, treating it as a ratio attribute.

Dataset: 48,842 records | 14 attributes | Target: Income >\$50K (binary)

Investigation: The model assumed equal intervals between education levels (High School to Some-College gap = College to Bachelors gap), which is incorrect. The ordinal attribute was also used directly in Euclidean distance for kNN, distorting neighbour selection.

Findings: Accuracy dropped from 85.1% (correct encoding) to 81.3% (raw integer). Confusion matrix showed poor recall for the >\$50K class.

Correct Approach: education-num was re-treated as ordinal and binned into 4 groups: {No School, K-12, Undergraduate, Graduate}. One-Hot Encoded. Model accuracy recovered to 85.4%.

Conclusion: Misclassifying ordinal attributes as ratio types introduces systematic bias and degrades model performance. Proper attribute profiling is essential before any modelling step.

9. Scenario-Based Learning

SCENARIO 1

SCENARIO 1: High-Cardinality Nominal Attribute in E-Commerce

Background: A data engineer is building a product recommendation model for an e-commerce platform with 500,000 unique product IDs.

Technical Environment: Python/pandas, scikit-learn, XGBoost, 2M transaction records.

Problem Statement: The "product_id" column (nominal, 500K unique values) was One-Hot Encoded, resulting in a 500,000-column sparse matrix that exceeded available RAM.

Available Evidence: `df.nunique()` = 500,000 for product_id. OHE matrix: $2M \times 500K = 1$ TB.

Investigation Approach: Identify the cardinality issue and select an alternative encoding.

Analysis: One-Hot Encoding is inappropriate for high-cardinality nominal attributes. Target Encoding, Hash Encoding, or Embedding Layers are needed.

Findings: Target Encoding reduced dimensionality to 1 column (mean income per product). XGBoost accuracy: 91.2%.

Recommended Solution: Apply Target Encoding (replace category with mean target value) or use an Embedding layer in a neural network (maps to 50-dim vector). Add k-fold smoothing to prevent leakage.

SCENARIO 2

SCENARIO 2: ZIP Code Misclassified as Ratio Attribute

Background: A junior data scientist in a real-estate ML project used ZIP codes (e.g., 600001, 600002) as integer features in a Linear Regression model for house price prediction.

Technical Environment: scikit-learn LinearRegression, pandas, 50,000 property records.

Problem Statement: The model produced negative house price predictions for some ZIP codes and showed poor $R^2 = 0.31$.

Available Evidence: ZIP code range: 600001–600100. Pearson correlation of ZIP vs Price: 0.02 (near zero).

Investigation Approach: Examine whether ZIP code values have ratio-scale meaning.

Analysis: ZIP codes are nominal identifiers (the number 600100 is not "larger" or "more" than 600001 in any meaningful sense). Using them as integers implies a false linear relationship.

Findings: Converting ZIP to One-Hot (or using Target Encoding with mean price per ZIP) improved R^2 from 0.31 to 0.74.

Recommended Solution: Always profile geographic codes (ZIP, PIN, postal code) as nominal attributes. Never use raw integer values of IDs as features.

SCENARIO 3

SCENARIO 3: Ordinal Encoding Ambiguity in Healthcare Survey

Background: A hospital analytics team built a pain level predictor using survey data where patients rated pain on a scale: None < Mild < Moderate < Severe < Extreme.

Technical Environment: R, caret package, ordinal logistic regression, 10,000 patient records.

Problem Statement: Using Label Encoding {None=0, Mild=1, Moderate=2, Severe=3, Extreme=4} in a standard linear model assumed equal intervals between pain levels, producing biased coefficient estimates.

Available Evidence: CLMM (cumulative link mixed model) diagnostics showed proportional odds assumption violated with raw integer encoding.

Investigation Approach: Compare standard label encoding vs Ordinal-specific modelling approaches.

Analysis: Standard regression with label-encoded ordinal targets misestimates probabilities because it assumes equal spacing between categories.

Findings: Using ordinal logistic regression (polr() in R) with the proper ordinal target produced correct confidence intervals and a 12% improvement in McFadden R².

Recommended Solution: Use ordinal logistic regression or threshold-based models for ordinal targets. Alternatively, apply SMOTE with ordinal-aware distance metrics for imbalanced ordinal classes.

10. Common Errors and Troubleshooting

Error / Mistake	Root Cause	Impact	Fix
Treating ZIP/ID as numeric ratio	Auto-inferred as int64 by pandas	False linear relationships in model	Cast to string/category before encoding
OHE on high-cardinality column	Applying OHE without checking nunique()	Memory overflow, sparse matrix	Use Target Encoding or Hashing Trick
Ordinal as interval in arithmetic	Assuming equal gaps in survey scales	Biased regression coefficients	Use ordinal regression or threshold models
Binary encoded as 0/1 integer	Label encoding symmetric binary attrs	May be correct but needs verification	Verify symmetric vs asymmetric; use SMC or Jaccard accordingly
Continuous variable forced discrete	Binning without domain knowledge	Loss of information, binning bias	Use equal-width or equal-frequency bins; verify with KDE

11. Key Terminologies

Term	Definition
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Data Object	A single instance/record in a dataset representing one entity (row in a table)
Attribute	A variable/feature describing a property of a data object (column in a table)
Nominal	Categorical attribute with no inherent order (e.g., colour, country)
Ordinal	Attribute with meaningful order but unequal intervals (e.g., satisfaction rating)
Interval	Numeric attribute with equal intervals but no true zero (e.g., temperature °C)
Ratio	Numeric attribute with true zero and all arithmetic operations valid (e.g., weight)
Binary	Attribute with only two possible values: 0 and 1
One-Hot Encoding	Converts a nominal attribute into k binary columns (k = number of categories)
Label Encoding	Maps ordinal categories to integers preserving their order
Feature Vector	A numeric vector $x = (x_1, x_2, \dots, x_n)$ representing one data object
Cardinality	Number of unique values in a categorical attribute
Target Encoding	Replaces each category with the mean of the target variable for that category

12. Topic Summary

UNIT II – Topic 1

DATA OBJECTS AND ATTRIBUTE TYPES — SUMMARY

- Data Objects (instances/records) are described by Attributes (features/variables) forming the Feature Matrix.
- Attributes are classified by measurement scale: Nominal (categories, no order), Ordinal (ordered, unequal intervals), Interval (equal intervals, no true zero), and Ratio (true zero, all operations valid).
- Binary attributes are either Symmetric (both states equal importance, use SMC) or Asymmetric (one state more important, use Jaccard).
- Encoding: Nominal → One-Hot Encoding | Ordinal → Label/Ordinal Encoding | Numeric → Normalization/Scaling.
- High-cardinality nominal attributes must not be One-Hot Encoded — use Target Encoding or embeddings.
- Misclassifying attribute types (ZIP as ratio, ordinal as interval) introduces model bias and reduces accuracy.
- Tools: pandas (dtypes, get_dummies), scikit-learn (LabelEncoder, OHE), WEKA, SPSS, SAS.
- This profiling step is the foundation for all subsequent preprocessing: cleaning, normalization, reduction, and transformation.